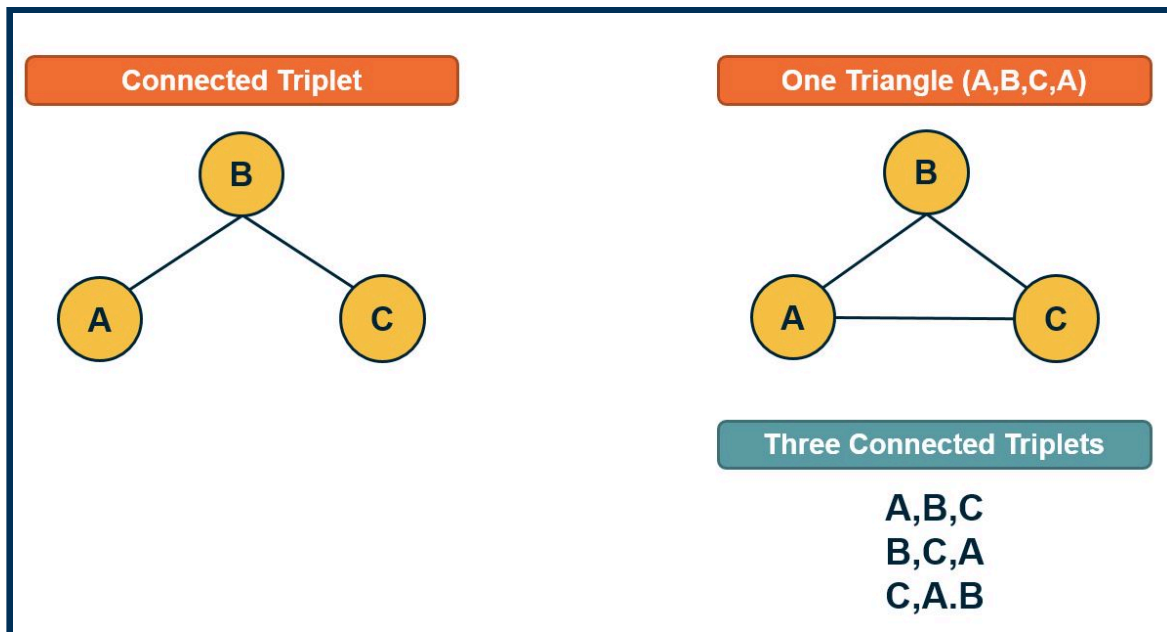


L5: Average Clustering and Transitivity Coefficient



If we want to quantify the degree of clustering in the entire network with a single number, we have two options. The first is to simply calculate the **Average Clustering Coefficient** for all nodes with degree larger than one.

A better metric, however, is to calculate the **Transitivity (or global clustering coefficient)**, which is defined as the **fraction of the connected triplets of nodes that form triangles**.

Mathematically, the transitivity is defined as:

$$T = \frac{3 \times \text{Number of triangles}}{\text{Number of connected triplets}}$$

A connected triplet is an ordered set of three nodes ABC such that A connects to B and B connects to C. For example, an A, B, C triangle corresponds to three triplets, ABC, BCA, and CAB. In contrast, a chain of connected nodes A, B, C, in which B connects to A and C, but A does not link to C, forms an open triplet ABC. The factor three in the previous equation is needed because each triangle is counted three times in the triplet count.

Food For Thought

Note that the Transitivity and the Average Clustering Coefficient are two different metrics. They may often be close but there are also some extreme cases in which the two metrics give very different answers. To see that consider a network in which two nodes A and B are connected to each other as well as to every other node. There are no other links. The total number of nodes is n . What would be the transitivity and average clustering coefficient in this case (*you can simplify by assuming that n is quite large*)?